Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01]

Circle geometry has three complementary languages: arc angles, point powers and circle equations. Choose one that preserves the given order of points. Directed-angle identities are powerful, but an ordinary angle or positive segment length still needs a branch check.

圆几何可用弧角、点幂及圆方程三种互补语言。选择能保留已知点序的方法。有向角恒等式虽然有力，但普通角及正线段长仍须筛选分支。

Chord lengths and arc angles share a common radius

弦长与弧角共享同一半径

Procedure

1. A chord subtending minor central angle θ has length $2R\sin(\theta/2)$.

2. An inscribed angle measures half the arc not containing its vertex; identify that arc from cyclic order.

3. For repeated chord types, add the actual consecutive central angles and exclude a hidden major-arc branch.

4. Use multiple-angle identities only after fixing those angle ranges.

通用步骤

1. 劣中心角为 θ 的弦长为 $2R\sin(\theta/2)$。

2. 圆周角等于不含其顶点的对应弧一半，应由圆上次序明确该弧。

3. 多次出现同类弦时，累加实际相邻中心角，并排除隐藏的优弧分支。

4. 确定角范围后，再用倍角或多角恒等式。

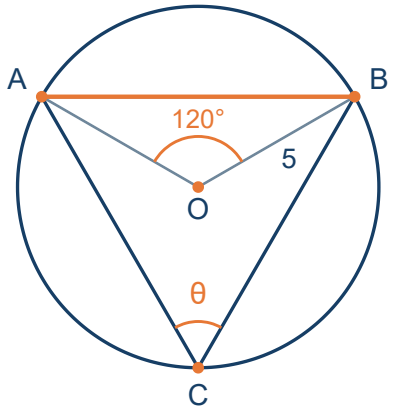
Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01] (continued / 续)

> Example. In a radius-5 circle a chord subtends 120°. Find its length and the inscribed angle subtending its minor arc.

> 例题：半径五的圆内，一弦所对劣中心角为 120°，求弦长及所对劣弧的圆周角。



Worked solution

1. The half central angle is 60°, so the chord is $10\sin 60^\circ = 5\sqrt{3}$.

2. An inscribed vertex outside this minor arc sees the 120° arc and has angle 60°.

3. A vertex on the minor arc instead sees the 240° complement and has angle 120°; the vertex location is essential.

例题解析

1. 中心半角为 60°，所以弦长为 $10\sin 60^\circ = 5\sqrt{3}$。

2. 劣弧外的圆周顶点所见对应弧为 120°，圆周角为 60°。

3. 若顶点在该劣弧上，则对应补弧为 240°，圆周角为 120°；顶点位置不可省略。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01] (continued / 续)

Equal chords determine equal minor arcs, but a consecutive arc could be the major complement.

等弦对应相等劣弧，但相邻顶点之间的实际弧可能是优弧补段。

A circular segment area is its sector area minus the center–chord triangle. Choose the smaller or complementary segment explicitly before forming a ratio.

圆弓形面积等于扇形减去圆心弦三角形；求比例前先明确使用小弓形还是补集。

Practice / 配套练习: AIME-2001-I-Q13 (WB p. 292), AIME-2004-II-Q01 (WB p. 239), AIME-2013-II-Q08 (WB p. 285), AIME-2015-I-Q06 (WB p. 286), AIME-2018-I-Q15 (WB p. 300), AIME-2025-II-Q05 (WB p. 282)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 2 / 方法 2 [GE04-M02]

Power multiplies directed secant distances

点幂使用有向割线距离乘积

Procedure

1. For a point outside a circle, $PT^2 = PA \cdot PB$ on a secant with near point A and far point B.

2. For intersecting chords at an interior point, the two positive endpoint-distance products are equal.

3. Converse equal products can create an auxiliary cyclic quadrilateral, provided all required points are distinct.

4. Use equal tangents to transfer lengths, then return to the actual segment order.

通用步骤

1. 圆外点满足 $PT^2 = PA \cdot PB$，其中 A 为近交点、B 为远交点。

2. 圆内弦交点到两端点的正距离乘积，在两弦上相等。

3. 反过来，等乘积可构造辅助共圆四边形，但须确认所需点互异。

4. 用等切线转移长度，再还原实际线段次序。

> Example. A secant from P meets a circle after distances $4, 9$. Another meets it first after distance three. Find its far-intersection distance and the tangent length.

> 例题：从 P 的一条割线在距离四、九处交圆，另一割线近交距为三，求远交距及切线长。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 2 / 方法 2 [GE04-M02] (continued / 续)

Worked solution

1. The power of P is $4 \cdot 9 = 36$.

2. If the second far distance is u, then $3u = 36$, giving $u = 12 > 3$.

3. The tangent length is $\sqrt{36} = 6$. Both positive-distance interpretations agree with P being outside.

例题解析

1. P 的点幂为 $4 \cdot 9 = 36$。

2. 第二条远交距 u 满足 $3u = 36$，所以 $u = 12 > 3$。

3. 切线长为 $\sqrt{36} = 6$，两组正距离均符合圆外位置。

Near distance times chord length is not the power; use near distance times full far distance.

近交距乘弦长不是点幂，应乘到远交点的全距。

Practice / 配套练习: AIME-2002-I-Q13 (WB p. 293), AIME-2005-I-Q15 (WB p. 294), AIME-2009-I-Q12 (WB p. 319), AIME-2016-I-Q06 (WB p. 316), AIME-2016-II-Q10 (WB p. 287), AIME-2019-II-Q15 (WB p. 304), AIME-2022-I-Q11 (WB p. 308), AIME-2024-I-Q10 (WB p. 289), AIME-2025-I-Q06 (WB p. 290), AIME-2026-II-Q10 (WB p. 291)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 3 / 方法 3 [GE04-M03]

Right angles identify diameter circles and antipodal pairs

直角识别直径圆与对径点

Procedure

1. A right angle subtending two fixed points places its vertex on their diameter circle.

2. The midpoint of a right triangle's hypotenuse is its circumcenter.

3. Use an antipodal point to replace a cyclic angle by a perpendicular or a line intersection.

4. An inscribed rectangle has two opposite-vertex pairs that are diameters.

通用步骤

1. 两固定点所张直角的顶点位于以两点连线为直径的圆上。

2. 直角三角形斜边中点即外心。

3. 利用对径点把圆内角换成垂直关系或直线求交。

4. 圆内接矩形的两组对顶点均互为对径点。

> Example. Points P, Q lie on the circle with diameter $AB = 10$ and subtend angle $PAQ = 30^\circ$. Find PQ.

> 例题：P, Q 在直径 $AB = 10$ 的圆上，且 $\angle PAQ = 30^\circ$，求 PQ。

Worked solution

1. The radius is five. The arc PQ not containing A has measure 60°.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 3 / 方法 3 [GE04-M03] (continued / 续)

2. The corresponding chord is $2 \cdot 5 \sin 30^\circ = 5$.

3. Thus $PQ = 5$; the same chord value follows if the minor central angle is taken directly as 60°.

例题解析

1. 圆半径为五，不含 A 的对应弧 PQ 为 60°。

2. 弦长为 $2 \cdot 5 \sin 30^\circ = 5$。

3. 所以 $PQ = 5$，亦可直接用 60° 劣中心角核实。

A midpoint locus needs a converse construction, not just one right-angle implication.

中点轨迹还需反向构造，不能只证一侧的直角推论。

Practice / 配套练习: AIME-2004-I-Q04 (WB p. 238), AIME-2011-II-Q10 (WB p. 284), AIME-2012-II-Q15 (WB p. 297), AIME-2014-I-Q15 (WB p. 298), AIME-2017-I-Q08 (WB p. 771), AIME-2020-II-Q15 (WB p. 305), AIME-2023-II-Q07 (WB p. 626), AIME-2025-I-Q14 (WB p. 190)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04]

Circle subtraction gives radical axes and their concurrence

两圆方程相减给出根轴及共线关系

Procedure

1. For circles with distinct centers, subtract their equations to cancel $x^2 + y^2$; the equal-power locus is the resulting radical-axis line, containing any common chord.

2. A point of equal powers lies on this radical axis even when its tangency segments are easier to compute than coordinates.

3. Three pairwise radical axes concur or are parallel, yielding auxiliary intersections and tangent-midpoint alignments.

4. Track the full ray order before turning directed power products into positive length products.

通用步骤

1. 对圆心不同的两圆，相减方程消去 $x^2 + y^2$；等点幂轨迹是所得根轴直线，若有公共弦则包含它。

2. 一点对两圆点幂相等便在根轴上；有时切线段比坐标更易求。

3. 三个两两根轴共点或平行，可得到辅助交点及公切线中点的共线关系。

4. 将有向点幂积转为正长度积之前，明确完整射线点序。

> Example. Find the common chord line of $x^2 + y^2 = 25$ and $(x-4)^2 + y^2 = 9$.

> 例题：求 $x^2 + y^2 = 25$ 与 $(x-4)^2 + y^2 = 9$ 的公共弦直线。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04] (continued / 续)

Worked solution

1. Expand the second equation as $x^2 + y^2 - 8x + 7 = 0$. The first is $x^2 + y^2 - 25 = 0$.

2. Subtract to get $-8x + 32 = 0$, so the radical axis is $x = 4$.

3. On that line $y^2 = 9$, giving distinct common points $(4, 3), (4, -3)$ and confirming a genuine chord.

例题解析

1. 第二式展开为 $x^2 + y^2 - 8x + 7 = 0$，第一式为 $x^2 + y^2 - 25 = 0$。

2. 相减得 $-8x + 32 = 0$，根轴为 $x = 4$。

3. 代入得 $y^2 = 9$，交点为 $(4, 3), (4, -3)$，确认是真正公共弦。

The radical axis is a line of equal powers, not necessarily a line of equal distances to centers.

根轴是等点幂直线，不一定是到两圆心等距直线。

The power difference has gradient magnitude $2O_1O_2$. Therefore its magnitude at P equals $2O_1O_2$ times distance from P to the radical axis. Along a center line, signed distances interpolate affinely.

The midpoint of a common external tangent segment has equal squared tangent powers, hence lies on the radical axis. Combine this with near/far intersection order before using a difference of squares.

点幂差的梯度长度为 $2O_1O_2$，所以一点幂差绝对值等于其到根轴距离乘 $2O_1O_2$。沿圆心直线，有向距离按仿射比例变化。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04] (continued / 续)

公外切线段中点的两切线幂相等，故位于根轴上。使用平方差之前，须结合近远交点的次序。

Practice / 配套练习: AIME-2005-II-Q08 (WB p. 332), AIME-2010-II-Q15 (WB p. 296), AIME-2016-I-Q15 (WB p. 299), AIME-2019-I-Q13 (WB p. 302), AIME-2019-I-Q15 (WB p. 303), AIME-2019-II-Q11 (WB p. 406), AIME-2021-I-Q13 (WB p. 348), AIME-2023-II-Q09 (WB p. 288)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 5 / 方法 5 [GE04-M05]

Circle equations linearize symmetric squared-distance sums

圆方程线性化对称距离平方和

Procedure

1. Choose a circle center or a distinguished vertex as origin.

2. For a point on a known circle, replace $x^2 + y^2$ by the circle equation's affine expression.

3. For circumcenter vector o with a vertex at zero, use $2o \cdot a = |a|^2$ for every other vertex vector.

4. A tangent can be written with a unit normal, giving exact projection coordinates and a bounded parameter range.

通用步骤

1. 以圆心或特殊顶点为原点。

2. 点在已知圆上时，将 $x^2 + y^2$ 替换为圆方程给出的仿射表达式。

3. 若一顶点在原点、外心向量为 o，其余顶点向量 a 满足 $2o \cdot a = |a|^2$。

4. 用单位法向量表示切线，可得到精确投影坐标及有界参数范围。

> Example. A point $P = (x, y)$ lies on $x^2 + y^2 = 4y$. Show $PA^2 + PB^2$ is affine in y when $A = (-1, 0), B = (1, 0)$.

> 例题：点 $P = (x, y)$ 在 $x^2 + y^2 = 4y$ 上，$A = (-1, 0), B = (1, 0)$，将 $PA^2 + PB^2$ 化为关于 y 的仿射式。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 5 / 方法 5 [GE04-M05] (continued / 续)

Worked solution

1. Expanding gives $PA^2 + PB^2 = 2(x^2 + y^2) + 2$.

2. The circle relation replaces $x^2 + y^2$ by $4y$.

3. Thus the sum equals $8y + 2$. This linearization avoids solving both point coordinates.

例题解析

1. 展开得 $PA^2 + PB^2 = 2(x^2 + y^2) + 2$。

2. 圆方程给出 $x^2 + y^2 = 4y$。

3. 所以平方和为 $8y + 2$，无须分别解出两坐标。

Use the circle equation only for points already proved to lie on that circle.

只有已经证明位于该圆上的点，才能代入其圆方程。

Practice / 配套练习: AIME-2001-II-Q06 (WB p. 283), AIME-2008-I-Q14 (WB p. 186), AIME-2015-II-Q11 (WB p. 402), AIME-2020-II-Q15 (WB p. 305), AIME-2023-II-Q12 (WB p. 412), AIME-2024-I-Q05 (WB p. 363)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 6 / 方法 6 [GE04-M06]

Directed circle angles still need ordinary-angle branch checks

圆内有向角仍需普通角分支检验

Procedure

1. Transfer angles through a shared chord or the tangent–chord theorem modulo 180°.

2. Use the converse cyclic-angle criterion to create useful auxiliary circles.

3. Recover the actual rays from convexity, cyclic point order, acute-angle bounds or explicit coordinates.

4. If a complex cross ratio is real, interpret equality of directed arguments as a generalized-circle condition and check excluded points.

通用步骤

1. 通过同弦或切线弦定理传递模 180° 的角。

2. 用共圆角判定的逆命题构造辅助圆。

3. 利用凸性、圆上点序、锐角界或明确坐标恢复实际射线。

4. 复交比为实时，将有向辐角相等解释为广义圆条件，并检查无定义点。

> Example. Points A, B, C, D are cyclic. If $\angle ABC = 70^\circ$ and B, D lie on opposite sides of AC, find $\angle ADC$.

> 例题：A, B, C, D 共圆，$\angle ABC = 70^\circ$，且 B, D 位于弦 AC 两侧，求 $\angle ADC$。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 6 / 方法 6 [GE04-M06] (continued / 续)

Worked solution

1. The vertices B, D see the two complementary arcs of AC.

2. Their ordinary inscribed angles sum to 180°, so $\angle ADC = 110^\circ$.

3. Directed equality modulo 180° remains true, but replacing it by ordinary-angle equality would incorrectly give seventy.

例题解析

1. B, D 分别看到 AC 的两条互补弧。

2. 两普通圆周角互补，所以 $\angle ADC = 110^\circ$。

3. 有向角模 180° 的相等仍成立，但不能误写为普通角也相等。

A supplementary branch can satisfy every squared length equation.

补角分支可能满足所有距离平方方程。

For noncollinear fixed complex points z_1, z_2, z_3, reality of $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ gives their circumcircle, excluding denominator-zero points. An ordinary angle must still be chosen from the actual rays.

对不共线定复点 z_1, z_2, z_3，$[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ 为实数时，轨迹为其外接圆，须排除分母为零的点。普通角仍需按实际射线筛选。

Practice / 配套练习: AIME-2012-II-Q15 (WB p. 297), AIME-2014-II-Q14 (WB p. 322), AIME-2015-I-Q06 (WB p. 286), AIME-2016-I-Q15 (WB p. 299), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q13 (WB p. 302), AIME-2021-II-Q14 (WB p. 307)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07]

A near incircle tangent gives a reciprocal side identity

内切圆近切线给出边长倒数恒等式

Procedure

1. Let the incircle touch AB, AC, BC at Z, W, X. Let Y be the first circle intersection on ray AX, and let its tangent cut side segments AB, AC at P, Q. Set $\ell = AZ = AW, p = AP, q = AQ$.

2. Equal tangents give $PY = \ell - p$ and $BX = AB - \ell$. The common angle at A and supplementary tangent–chord angles give the sine-rule ratio $PY/p = BX/AB$.

3. Rearrange to $\ell/p + \ell/AB = 2$. Repeating on the other side gives $\ell/q + \ell/AC = 2$.

4. Check the near-intersection order and positive cut segments. An arbitrary tangent to the near incircle arc does not satisfy these reciprocal laws without A, Y, X collinear.

通用步骤

1. 设内切圆切 AB, AC, BC 于 Z, W, X。令 Y 为射线 AX 先遇到的圆交点，Y 处切线交边线段 AB, AC 于 P, Q；记 $\ell = AZ = AW, p = AP, q = AQ$。

2. 等切线给出 $PY = \ell - p, BX = AB - \ell$。由共同的 A 角及互补的弦切角，正弦定理得 $PY/p = BX/AB$。

3. 整理得 $\ell/p + \ell/AB = 2$，另一边同理得 $\ell/q + \ell/AC = 2$。

4. 检查近交点次序及各截段为正。若缺少 A, Y, X 共线条件，任意近弧切线并不满足这些倒数关系。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07] (continued / 续)

> Example. The incircle of ABC touches BC at X. Let Y be the first intersection of ray AX with the incircle, and let the tangent at Y meet AB, AC at P, Q inside the sides. Prove $AP = AQ$ if and only if $AB = AC$.

> 例题：三角形内切圆切 BC 于 X。射线 AX 先交圆于 Y，Y 处切线交边内于 $P \in AB, Q \in AC$。证明 $AP = AQ$ 当且仅当 $AB = AC$。

Worked solution

1. Write $p = AP, q = AQ$ and let ℓ be the common vertex tangent length. The tangent–chord and sine-rule calculation gives $1/p + 1/AB = 2/\ell$.

2. The same argument on the AC side gives $1/q + 1/AC = 2/\ell$. Subtracting yields $1/p - 1/q = 1/AC - 1/AB$.

3. All four lengths are positive. Thus the left side vanishes exactly when $p = q$, and the right side exactly when $AB = AC$, proving both directions.

例题解析

1. 记 $p = AP, q = AQ$，公共顶点切线长为 ℓ。由弦切角与正弦定理得到 $1/p + 1/AB = 2/\ell$。

2. 在 AC 一侧同理得到 $1/q + 1/AC = 2/\ell$。相减得 $1/p - 1/q = 1/AC - 1/AB$。

3. 四个长度均为正，所以左边为零等价于 $p = q$，右边为零等价于 $AB = AC$，两方向均得证。

The far incircle tangent does not have the same segment signs.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07] (continued / 续)

内切圆远切线的线段符号并不相同。

Practice / 配套练习: AIME-2018-II-Q14 (WB p. 301)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 8 / 方法 8 [GE04-M08]

Cyclic intercept ratios are products of chord ratios

圆内截比等于弦比乘积

Procedure

1. For a cyclic quadrilateral, split triangles at a chord intersection and use common-altitude area ratios.

2. Apply the sine rule to the corresponding equal inscribed angles; products of chord lengths replace unknown sines.

3. Convert a division ratio t on a unit segment to position $t/(1+t)$ before subtracting positions.

4. Check that the free point traverses the complete positive parameter range needed for equality.

通用步骤

1. 对圆内接四边形，在弦交点分三角形并使用等高面积比。

2. 对相等圆周角使用正弦定理，用弦长乘积替换未知正弦。

3. 单位线段上分比为 t 时，其位置是 $t/(1+t)$，先换位置再相减。

4. 检查活动点是否遍历等号所需的全部正参数范围。

> Example. On a unit diameter, two chord-intersection points have ratios from the left endpoint x and $2x$, with $x>0$. Find their largest separation.

> 例题：在单位直径上，两弦交点从左端量起的分比分别为 $x, 2x$，其中 $x>0$，求两点最大间距。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 8 / 方法 8 [GE04-M08] (continued / 续)

Worked solution

1. The positions are $x/(1+x)$ and $2x/(1+2x)$, so their difference is $x/((1+x)(1+2x))$.

2. Its reciprocal is $1/x+3+2x \geq 3+2\sqrt{2}$, with equality at $x=1/\sqrt{2}$.

3. The largest separation is $1/(3+2\sqrt{2}) = 3-2\sqrt{2}$.

例题解析

1. 两位置为 $x/(1+x)$ 与 $2x/(1+2x)$，差为 $x/((1+x)(1+2x))$。

2. 倒数为 $1/x+3+2x \geq 3+2\sqrt{2}$，在 $x=1/\sqrt{2}$ 取等。

3. 故最大间距为 $3-2\sqrt{2}$。

For diameter endpoints M, N and cyclic B, C, $MP/PN = MB \cdot MC/(NB \cdot NC)$ at $P = BC \cap MN$.

直径端点为 M, N、圆上点为 B, C 时，$P = BC \cap MN$ 满足 $MP/PN = MB \cdot MC/(NB \cdot NC)$。

Practice / 配套练习: AIME-2009-II-Q15 (WB p. 187)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 9 / 方法 9 [GE04-M09]

Cyclic quadrilaterals combine Ptolemy, area and circumradius

圆内接四边形结合托勒密、面积与外接半径

Procedure

1. For consecutive sides a, b, c, d, use Ptolemy: the diagonal product equals $ac + bd$.

2. Brahmagupta gives $K^2 = (s-a)(s-b)(s-c)(s-d)$.

3. If two pieces share a circumcircle, equate $16K^2R^2 = (ab+cd)(ac+bd)(ad+bc)$.

4. Reject algebraic roots violating positive side lengths, convexity or the cyclic point order.

通用步骤

1. 连续四边为 a, b, c, d 时，托勒密给出对角线乘积为 $ac + bd$。

2. 婆罗摩笈多公式为 $K^2 = (s-a)(s-b)(s-c)(s-d)$。

3. 两块共外接圆时，可比较 $16K^2R^2 = (ab+cd)(ac+bd)(ad+bc)$。

4. 排除违反正边长、凸性或圆上次序的代数根。

> Example. A cyclic quadrilateral has consecutive sides $3, 4, 3, 4$. Find its area and diagonal product.

> 例题：圆内接四边形连续边长为三、四、三、四，求面积及对角线乘积。

Worked solution

1. The semiperimeter is seven, so $K^2 = 4 \cdot 3 \cdot 4 \cdot 3 = 144$ and $K = 12$.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 9 / 方法 9 [GE04-M09] (continued / 续)

2. Ptolemy gives diagonal product $3 \cdot 3 + 4 \cdot 4 = 25$.

3. It is a rectangle with diagonals five; the direct 3×4 area checks both formulas.

例题解析

1. 半周长为七，所以 $K^2 = 4 \cdot 3 \cdot 4 \cdot 3 = 144$，面积为十二。

2. 托勒密给出对角线乘积 $3 \cdot 3 + 4 \cdot 4 = 25$。

3. 该构型为三乘四矩形、对角线五，直接计算核实两式。

Brahmagupta’s short formula needs cyclicity; a general quadrilateral has an extra angle term.

婆罗摩笈多简式需要共圆，一般四边形还含额外角度项。

Cross-projection onto two intersecting diagonals is reflection across an angle bisector followed by scale $\cos\theta$. Hence all projected side lengths scale by that factor, while $K = \frac{1}{2} d_1 d_2 \sin\theta$ determines it from Ptolemy and area.

向两相交对角线作交叉投影，等价于关于角平分线反射再按 $\cos\theta$ 缩放。投影边长均按此比例变化，而 $K = \frac{1}{2} d_1 d_2 \sin\theta$ 可结合托勒密与面积求该比例。

Practice / 配套练习: AIME-2021-I-Q11 (WB p. 306), AIME-2022-II-Q15 (WB p. 309)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 10 / 方法 10 [GE04-M10]

Arc and semicircle structure controls circular counting

圆弧及半圆结构控制圆周计数

Procedure

1. An inscribed obtuse triangle has all three vertices in one open semicircle; choose a unique first vertex when counting.

2. Distinct circles intersect in at most two points; coincident boundary circles give no proper strictly separated subset. Thus a disk selects one consecutive run of vertices on a circle, unless empty or full.

3. A random chord misses a fixed inscribed convex polygon exactly when both endpoints lie in the same open boundary arc between consecutive vertices.

通用步骤

1. 圆内接钝角三角形的三顶点位于同一开半圆；计数时可指定唯一首顶点。

2. 不同圆至多交于两点；边界圆重合时不能严格分离非空真子集。因此圆盘在圆周顶点中选出的子集应为连续一段，或空集、全集。

3. 随机弦不交固定圆内接凸多边形，当且仅当两端点位于同一对相邻顶点之间的开圆弧。

> Example. A fixed inscribed equilateral triangle is given. Two independent uniform circumference points form a chord. Find the probability the chord meets the triangle.

> 例题：给定圆内接等边三角形，从圆周独立均匀取两点连弦，求弦与三角形相交的概率。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 10 / 方法 10 [GE04-M10] (continued / 续)

Worked solution

1. The three complementary boundary arcs each have one third of the circumference.

2. The chord misses the triangle precisely when both endpoints belong to the same one of these arcs, apart from zero-probability endpoint cases.

3. The miss probability is $3(1/3)^2 = 1/3$, so the intersection probability is $2/3$.

例题解析

1. 三条相邻顶点间的边界圆弧各占圆周三分之一。

2. 除概率零的端点情况外，弦不交三角形恰在两端位于同一此类圆弧时发生。

3. 不交概率为 $3(1/3)^2 = 1/3$，故相交概率为 $2/3$。

An open semicircle excludes antipodal right-angle cases.

开半圆不包含对径点对应的直角情形。

Practice / 配套练习: AIME-2011-I-Q10 (WB p. 739), AIME-2014-I-Q05 (WB p. 693), AIME-2017-I-Q06 (WB p. 770)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 11 / 方法 11 [GE04-M11]

The orthocenter reflects a circumcircle across a side

垂心把外接圆关于一边反射

Procedure

1. Assume the orthocenter H and B, C are distinct. With the circumcenter at the vector origin, use $h = a + b + c$.

2. The circle through H, B, C is the reflection of the original circumcircle across BC.

3. Its center is $b + c = h - a$; the radius toward H is parallel to a.

4. Use chord midpoint normals or a tangent perpendicular to that radius to recover lengths and area.

通用步骤

1. 先假设垂心 H 与 B, C 互不重合。以外心为向量原点，使用 $h = a + b + c$。

2. 过 H, B, C 的圆是原外接圆关于 BC 的反射。

3. 其圆心为 $b + c = h - a$，指向 H 的半径平行于 a。

4. 利用弦中点法向或垂直该半径的切线，求长度与面积。

> Example. On the unit circle take $A = (0, 1), B = (-1, 0), C = (1, 0)$. Find the center of the circle through the orthocenter H and B, C.

> 例题：单位圆上取 $A = (0, 1), B = (-1, 0), C = (1, 0)$，求过垂心 H 与 B, C 的圆心。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 11 / 方法 11 [GE04-M11] (continued / 续)

Worked solution

1. Triangle ABC is right at A, so $H = A = (0, 1)$; the vector identity also gives $a + b + c = a$.

2. The side BC is the x axis, so reflection preserves the original circle in this special case.

3. The new center is $b + c = (0, 0)$, and its radius is one.

例题解析

1. 三角形在 A 为直角，故 $H = A = (0, 1)$，向量恒等式也给出 $a + b + c = a$。

2. BC 为 x 轴，在此特殊情况下反射保持原圆不变。

3. 新圆心为 $b + c = (0, 0)$，半径为一。

The reflected circle need not have the same center as the original in a general acute triangle.

一般锐角三角形中，反射后的圆心不等于原圆心。

Practice / 配套练习: AIME-2020-I-Q15 (WB p. 408)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 12 / 方法 12 [GE04-M12]

A fixed inscribed angle gives a circular arc with an attainable height bound

固定圆周角给出圆弧及可达高度界

Procedure

1. Use cyclic angle relations to show a moving point sees a fixed chord under a fixed ordinary angle.

2. This places it on a specified arc; height above the chord is greatest at the arc midpoint.

3. Prove the midpoint is attainable using an explicit configuration or a continuous parameter with distinct endpoint limits.

通用步骤

1. 由圆周角关系证明活动点看固定弦的普通角恒定。

2. 因此活动点位于指定圆弧，距弦高度在弧中点最大。

3. 用明确构造或具有不同端点极限的连续参数，证明弧中点确实可达。

> Example. Point P lies on the minor circular arc below a chord $BC = 2$ and satisfies $\angle BPC = 120^\circ$. Find the largest triangle area.

> 例题：点 P 位于弦 $BC = 2$ 下方的劣圆弧上，且 $\angle BPC = 120^\circ$，求三角形面积最大值。

Worked solution

1. At the maximum $PB = PC = x$. Cosine law gives $4 = 2x^2 - 2x^2 \cos 120^\circ = 3x^2$.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 12 / 方法 12 [GE04-M12] (continued / 续)

2. The corresponding height is $x\cos 60^\circ = 1/\sqrt{3}$, reached at the arc midpoint.

3. The area is $2(1/\sqrt{3})/2 = 1/\sqrt{3}$.

例题解析

1. 最大时 $PB = PC = x$，余弦定理给出 $4 = 3x^2$。

2. 相应高为 $x\cos 60^\circ = 1/\sqrt{3}$，在弧中点取到。

3. 面积为 $2(1/\sqrt{3})/2 = 1/\sqrt{3}$。

A locus bound is insufficient if the moving configuration reaches only part of the arc.

若原构型只能到达圆弧的一部分，仅有轨迹上的上界还不充分。

Practice / 配套练习: AIME-2009-I-Q15 (WB p. 295), AIME-2013-II-Q10 (WB p. 252)